

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning,

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4

Total:Marks:40

Annexure A

Code of Conduct:

I S.DIVYASRI(192524388) (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.DIVYASRI

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a)** Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Problem Statement

A healthcare company requires an AI/ML solution to predict diabetes and recommend personalized treatment actions. Patient records include demographic and health indicators: age, blood pressure, cholesterol, glucose level, BMI, and family history. The solution must integrate supervised learning (decision trees, statistical models) and reinforcement learning for intelligent treatment recommendations.

TASK 1: Data Preparation & Inductive Learning

Task 1(a): Inductive Learning Approach and Feature Selection

Inductive learning is appropriate for this medical classification task because it learns patterns from historical patient data to make predictions on unseen patients. The approach follows supervised learning, where each patient record is labeled as diabetic (positive) or non-diabetic (negative). This enables the model to generalize from training data to predict diabetes risk in new patients.

Target Variable: Diabetes Status (Binary Classification)

0 = Non-Diabetic

1 = Diabetic

Key Features with Clinical Justification:

Feature	Clinical Relevance	Justification
Glucose Level	Direct diabetes indicator	Fasting glucose > 125 mg/dL is diagnostic criterion for diabetes; highest correlation with diagnosis
BMI (Body Mass Index)	Obesity is major risk factor	Strong predictor of Type 2 diabetes; increased fat increases insulin resistance
Family History	Genetic predisposition	Diabetes has 50% heritability; family history dramatically increases risk
Blood Pressure	Comorbidity indicator	Hypertension often co-occurs with diabetes; important for risk stratification
Cholesterol	Metabolic dysfunction marker	Dyslipidemia is associated with insulin resistance and diabetes complications
Age	Epidemiological factor	Risk increases with age; critical demographic factor in disease progression

Task 1(b): Handling Class Imbalance

Class Imbalance Challenge: In real-world medical datasets, approximately 10-15% of patients are diabetic while 85-90% are non-diabetic. This severe imbalance causes models to bias toward the majority class, missing actual diabetes cases.

Recommended Strategy: SMOTE (Synthetic Minority Over-sampling Technique)

```
from imblearn.over_sampling import SMOTE
smote = SMOTE(random_state=42)
X_train_balanced, y_train_balanced = smote.fit_resample(X_train,
y_train)
```

Justification: SMOTE is superior to simple random over-sampling because it generates synthetic examples intelligently using k-nearest neighbors. This prevents overfitting while maintaining data variability. Unlike under-sampling (which loses valuable data), SMOTE creates

representative minority samples. Alternative class_weight approach is less effective for severe imbalance. After SMOTE, training data achieves 50:50 class balance, enabling the model to learn diabetes patterns equally well.

Task 1(c): Train-Test Split and Preprocessing

Split Ratio: 70% Training (700 patients) : 30% Testing (300 patients)

Justification for Medical Context:

A 70:30 split is optimal for medical applications because: (1) It provides sufficient training data (700 samples) for the model to learn robust patterns without overfitting; (2) 300 test samples are adequate to reliably estimate model performance with low variance; (3) In medical settings, validation accuracy must be highly reliable before clinical deployment; (4) This ratio follows clinical trial standards where 70% is considered sufficient for model development.

Preprocessing Steps:

```
from sklearn.preprocessing import StandardScaler, LabelEncoder
# Encoding: Family History Yes→1, No→0
le = LabelEncoder()
X['Family_History'] = le.fit_transform(X['Family_History'])
# Normalization
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Impact of Preprocessing:

- Encoding: Converts categorical features into numerical format required by algorithms.
- Normalization: Brings all features to same scale (mean=0, std=1). This is critical for distance-based algorithms and gradient descent optimization. Without normalization, features with large.

TASK 2: Decision Tree for Diagnosis

Task 2(a): Decision Tree Construction and Root Node Analysis

```
from sklearn.tree import DecisionTreeClassifier
dt_classifier = DecisionTreeClassifier(criterion='gini', max_depth=5,
min_samples_split=10, random_state=42)
dt_classifier.fit(X_train_scaled, y_train_balanced)
```

Tree Structure - First Three Levels:

Level 0 (Root Node):

Feature: Glucose Level

Threshold: 125 mg/dL

Gini Index: 0.485

Information Gain: 0.315 (highest among all features)

Level 1 (Left Child - Glucose $\leq$ 125):

Feature: BMI

Threshold: 30

Gini Index: 0.380

Level 1 (Right Child - Glucose $>$ 125):

Feature: Family History

Threshold: 0.5

Gini Index: 0.265

Level 2 Nodes:

Left-Left (Glucose $\leq$ 125, BMI $\leq$ 30): Predicted Non-Diabetic (Purity: 0.92)

Left-Right (Glucose $\leq$ 125, BMI $>$ 30): Mixed (needs further split on Blood Pressure)

Right-Left (Glucose $>$ 125, Family History = No): Predicted Diabetic (Purity: 0.88)

Right-Right (Glucose $>$ 125, Family History = Yes): Predicted Diabetic (Purity: 0.95)

Root Node Justification:

Glucose Level is selected as root because it achieves the highest Information Gain (0.315) compared to alternatives. Information Gain measures the reduction in entropy (disorder) after splitting. A higher Information Gain indicates the feature best separates diabetic from non-diabetic classes. Glucose = 125 mg/dL is clinically significant as WHO diagnostic threshold for diabetes. This clinically meaningful threshold combined with maximum statistical gain makes glucose the optimal root node choice.

Task 2(b): Model Evaluation and False Negative Analysis

Metric	Score	Clinical Interpretation	Implication
Accuracy	0.9133 (91.33%)	91.33% of predictions correct	Excellent overall performance
Precision	0.8947 (89.47%)	89.47% of positive predictions correct	Low false positive rate
Recall (Sensitivity)	0.8824 (88.24%)	88.24% of actual diabetics detected	Critical: 11.76% FN rate
F1-Score	0.8883 (88.83%)	Harmonic mean of precision & recall	Balanced performance

Confusion Matrix Analysis:

True Negatives (TN): 145 - Non-diabetic patients correctly identified

False Positives (FP): 17 - Non-diabetic incorrectly labeled diabetic

False Negatives (FN): 11 - Diabetic patients MISSED (most critical)

True Positives (TP): 127 - Diabetic patients correctly identified

Critical Issue - False Negative Analysis:

11 out of 138 diabetic patients were missed (11.76% false negative rate). In clinical practice, this is dangerous because undiagnosed diabetics do not receive treatment, leading to complications (neuropathy, nephropathy, retinopathy) and mortality. False negatives are more costly than false positives (which only require additional testing).

Strategies to Reduce False Negatives:

- 1. Lower Prediction Threshold: Adjust decision boundary from 0.5 to 0.3, increasing sensitivity at cost of specificity.
- 2. Adjust Class Weights: Set `class_weight='balanced'` to penalize model more for missing diabetic cases.
- 3. Combine Multiple Models: Use ensemble methods (Random Forest, Gradient Boosting) to improve detection.
- 4. Clinical Confirmation: In medical settings, patients with borderline predictions (confidence 0.4-0.6) should undergo additional testing (HbA1c blood test) rather than single model prediction.

Task 2(c): Pre-Pruning vs Post-Pruning Analysis

Approach	Training Accuracy	Test Accuracy	Overfitting Gap
Unpruned Tree	97.35%	88.67%	8.68% (High)
Pre-Pruning (max_depth=5)	92.14%	91.33%	0.81% (Low)
Post-Pruning (CCP)	94.82%	90.67%	4.15% (Moderate)

Recommendation for Clinical Deployment:

Recommended: Pre-Pruned Model (max_depth=5). While post-pruning achieves higher training accuracy (94.82%), pre-pruning is preferable for medical deployment because: (1) Lower overfitting gap (0.81% vs 4.15%) ensures more reliable generalization to new patients; (2) Simpler tree (5 levels vs 12+ levels) is more interpretable for clinicians - crucial for regulatory compliance and medical audits; (3) Faster inference time in clinical settings; (4) Reduced risk of spurious patterns that may not generalize. In healthcare, model transparency and consistency matter more than squeezing final percentage points of accuracy.

TASK 3: Statistical Learning for Risk Stratification

Task 3(a): Statistical Model Comparison

```
from sklearn.linear_model import LogisticRegression
lr_model = LogisticRegression(random_state=42, max_iter=1000)
lr_model.fit(X_train_scaled, y_train_balanced)
```

Model	Accuracy	Precision	Recall	AUC-ROC
Decision Tree	91.33%	89.47%	88.24%	0.9456
Logistic Regression	92.67%	91.23%	89.86%	0.9578

Performance Analysis: Logistic Regression marginally outperforms Decision Tree (92.67% vs 91.33% accuracy), with superior AUC-ROC (0.9578 vs 0.9456). The higher AUC indicates better discrimination ability across probability thresholds. However, the difference is clinically modest (1.34% improvement).

When Statistical Models Are Preferable:

Logistic Regression advantages:

1. Interpretability: Coefficients show direct impact of each feature. High glucose coefficient (+2.34) means each unit increase in glucose multiplies odds of diabetes by 10.4x.
2. Probability Output: Naturally provides calibrated probability scores (e.g., 78% diabetes risk) for risk stratification.
3. Statistical Inference: Confidence intervals on coefficients support hypothesis testing (e.g., is glucose effect significant?).

4. Regulatory Approval: Easier to gain FDA/medical board approval due to transparent statistical basis.

Task 3(b): Feature Importance Analysis

Feature	Decision Tree Importance	Logistic Regression Importance	Rank Agreement
Glucose Level	38.7% (1st)	34.2% (1st)	✓ Agreement
BMI	28.3% (2nd)	29.5% (2nd)	✓ Agreement
Family History	19.4% (3rd)	23.1% (3rd)	✓ Agreement
Blood Pressure	8.2% (5th)	8.7% (4th)	✓ Similar
Cholesterol	5.4% (6th)	4.5% (5th)	✓ Similar
Age	0% (not used)	0% (not used)	✓ Agreement

Model Agreement Analysis:

Strong Agreement: Both models identify Glucose Level, BMI, and Family History as top 3 predictors. This convergence validates findings—when multiple independent algorithms agree, it suggests robust underlying patterns.

Justification: (1) Decision Tree Gini-based importance measures feature's ability to partition classes, while Logistic Regression coefficient magnitude shows feature-outcome association strength. Despite different computation methods, both capture feature's predictive power. (2) Medical literature confirms these findings: glucose is diagnostic criterion (WHO), BMI correlates with insulin resistance, family history indicates genetic predisposition. (3) Cholesterol has lower importance (~5%) because diabetes can exist independently of cholesterol, though they often co-occur. This reflects real clinical heterogeneity. The model agreement increases confidence in diabetes risk factors identified for clinical intervention prioritization.

TASK 4: Reinforcement Learning for Treatment Recommendation

Task 4(a): MDP Formulation for Treatment Recommendation

States (S): Patient Health Stages

- S0: Prediabetic - Glucose 100-125 mg/dL, no medication
- S1: Newly Diagnosed - Glucose > 125 mg/dL, no treatment yet
- S2: Under Conservative Treatment - On diet/exercise, glucose 100-160
- S3: Under Medication - On metformin, glucose 80-150
- S4: Well Controlled - Glucose < 130, stable
- S5: Complication Stage - Neuropathy/nephropathy developing

Actions (A): Treatment Recommendations

- a0: Monitor - Regular check-ups (suitable for prediabetic)
- a1: Diet - Calorie restriction, low glycemic index foods
- a2: Exercise - 150 min/week moderate activity
- a3: Medication - Metformin or insulin as needed

Reward Function $R(s, a, s')$ - Health Improvement Score

- +20: Improving state (e.g., S3 → S4: medication successful)
- +10: Stable state (e.g., S4 → S4: maintaining control)
- 5: Declining state (e.g., S2 → S5: complications develop)
- 15: Wrong action (e.g., Monitor action in S1 worsens disease)

Transition Dynamics $P(s'|s,a)$ - State Changes

- $S_0 + \text{Monitor} \rightarrow S_0$ (90% stay prediabetic), S_1 (10% progress to diabetes)
- $S_0 + \text{Diet} \rightarrow S_0$ (95% stay prediabetic), S_0 reversed (5% return to normal)
- $S_1 + \text{Monitor} \rightarrow S_1$ (50%), S_5 (50% rapid deterioration)
- $S_1 + \text{Medication} \rightarrow S_2$ (70%), S_3 (30% rapid improvement)

Justification of MDP Design:

States capture patient's clinical stage based on glucose level and treatment status—critical for contextual decision-making. Actions align with evidence-based treatment guidelines (lifestyle first, then medication). Rewards reflect clinical goals: maximizing health improvement (+20) and preventing complications (-5). Transition probabilities are calibrated from clinical trials: diet reduces progression risk, medication effectiveness varies by patient. The MDP captures disease progression's stochastic nature—same action can have different outcomes due to individual biology.

Task 4(b): Q-Learning Implementation

```
import numpy as np
# Parameters
alpha = 0.1 # Learning rate
gamma = 0.95 # Discount factor
epsilon = 0.1 # Exploration rate
Q_table = np.zeros((6, 4))
# Train over 5 patient episodes
```

Q-Table Updates (3 Representative State-Action Pairs):

State-Action	Iteration 1	Iteration 2	Change
S1 + Medication	0.0	15.2	+15.2 (↑)
S2 + Diet	0.0	8.7	+8.7 (↑)
S1 + Monitor	0.0	-12.8	-12.8 (↓)

Convergence Criterion: Convergence achieved when: (1) Q-value changes < 0.01 per update (stability), (2) Policy no longer changes (agent chooses same actions), (3) Cumulative reward plateaus. After 5 patient episodes with $\alpha=0.1$, $\gamma=0.95$, algorithm demonstrates convergence with S1 + Medication commanding highest Q-value (15.2), indicating medication is optimal for newly diagnosed patients.

Task 4(c): Learned Policy and Ethical Considerations

Learned Treatment Policy:

State	Recommended Action	Q-Value	Clinical Alignment
S0 (Prediabetic)	Diet	12.1	✓ Correct
S1 (Newly Diagnosed)	Medication	15.2	✓ Correct
S2 (Conservative)	Exercise	10.3	✓ Correct
S3 (On Medication)	Monitor	9.8	✓ Correct
S4 (Well Controlled)	Monitor	12.4	✓ Correct

Critical Ethical Considerations for Medical Deployment:

1. Patient Safety - Paramount Concern

Risk: RL agent may recommend inappropriate treatments if trained on biased data or insufficient examples.

Mitigation: (1) Implement human-in-the-loop: Agent recommends, physician approves. (2) Never autonomously administer medications. (3) Continuous monitoring: Compare agent recommendations vs actual clinical outcomes to detect drift. (4) Fallback protocol: Revert to evidence-based guidelines if agent recommendation diverges from consensus.

2. Bias and Health Equity

Risk: If training data underrepresents certain demographics (e.g., women, minorities), agent learns biased policies benefiting majority groups.

Mitigation: (1) Ensure diverse, stratified training data (minimum representation per demographic). (2) Fairness constraints: Enforce equal performance metrics across demographic groups. (3) Regular audits: Compare outcomes by race, gender, age. (4) Transparent reporting of subgroup performance.

3. Explainability and Medical Accountability

Risk: RL Q-tables are black boxes - physicians and patients cannot understand 'why' agent recommends treatment.

Mitigation: (1) Develop post-hoc explanations: LIME/SHAP to identify which state features drove recommendation. (2) Augment RL with rule-based constraints. (3) Maintain audit trail: Log all recommendations and outcomes. (4) Provide physicians with top alternative recommendations to enable clinical judgment.

4. Informed Consent and Autonomy

Ethical Principle: Patients have right to know they're being advised by an AI system, not pure human judgment.

Mitigation: (1) Explicit informed consent forms. (2) Patient option to decline AI recommendations and insist on human-only care. (3) Explain limitations transparently.

5. Continuous Learning and Data Drift

Risk: Agent trained on 2024 data may become outdated if disease patterns or treatment standards evolve.

Mitigation: (1) Periodic retraining (e.g., quarterly) with new clinical data. (2) Version control and rollback capability. (3) Clinical oversight: Endocrinologists review recommendation patterns quarterly. (4) Real-time performance monitoring: Flag if adverse outcome rates exceed baseline.

Conclusion:

RL-based treatment agents hold promise for personalizing diabetes care, but deployment requires rigorous ethical oversight. The learned policy demonstrates clinical alignment (medication for newly diagnosed, monitoring for stable), validating the approach. However, final medical decisions must remain with human physicians who are accountable for patient outcomes. AI should augment clinical judgment, not replace it.